

***Report on***  
***App Arcade***

***Prepared for***

|                                                                                   |                                                                                |
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**Course No: CSE 3224**  
**Course Name: Information System Design & Software Engineering**  
**Lab**

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***Date: 23/6/2024***



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## **Summary of the Project**

This project aims to build a significant platform for both gamers and app developers called App Arcade. The platform holds both free and purchasable games and applications which a user can use for free or buy to use depending on the requirements. The objective of this project is to introduce a user friendly interface for different types of apps and games for all people.

## **Background Study**

Investigate gaming communities to gain insight into the preferences and expectations of gamers. It can study user behaviour also how user discovers apps and games. It can also gain experience from app developers by app distribution.

## **Feasibility Analysis**

Our project significantly contributes to social welfare through:

### **● Community Building and Social Interaction:**

The platform can serve as a hub for users to connect, share experiences, and engage in discussions about their favorite games and applications. This fosters a sense of community, reducing social isolation and promoting positive interactions.

### **● Educational Opportunities:**

Integrating educational games and applications within the platform can support learning and skill development.

### **● Support for Small Developers:**

By providing them with a platform to showcase their creations, the project contributes to economic empowerment and social inclusivity.

### ● Promotion of Mental Health:

Introducing features that promote mindfulness, stress reduction, or mental health awareness within the platform can contribute to users' overall well-being. It acknowledges the importance of mental health in the digital age.

By incorporating these elements into the Online Game and App Store project, it can go beyond being a mere content distribution platform and actively contribute to the social well-being of its user community.

### Requirement Analysis

- 1. User Authentication:** Verify user during login to authenticate users and grant access to the app arcade.
- 2. Product Catalog :** Implement search functionality,App/Game Listings.
- 3. Cart :** Cart Management
- 4.Payment Gateway :** Choose a payment gateway in some factors such as transaction,fees,security features.
- 5.Search and Filters :** User Ratings and Reviews , Search for apps and games , Showcase Featured Apps and Games.
- 6.User Library :** It provides user accounts who have accounts in library management system or can create accounts to access the user library feature.

### Risk Analysis

- 1. Cybersquatting Threats -** This app collects user data including personal information and payment information. A data leakage could result in sensitive information being stolen, leading to identity theft or financial loss. Accounts can be stolen easily because of weak authentication. The app may rely on third-party libraries which could be exploited by attackers to compromise the app security.
- 2. User Misuse -** User might upload inappropriate contents or share contents which could violate community guidelines and harm the apps reputation or user experience.
- 3. Technological Risk-** The apps servers or infrastructure may experience downtime due to hardware failures or cyber attacks resulting in service interruptions for users and potential financial losses.
- 4. Market Risk-** App Arcade may face intense market competition from other gaming platforms , app stores , which could limits its ability to attract and entertain users.
- 5. Operational Risk -** Operational Risks happen due to technical failures. Such as server outages , network issues , or software bugs could lead to service downtime , data loss or impacting user satisfaction and revenue generation.
- 6. Financial Risk -** Market competition , Financial Market risks are the major issues of Financial Risk.

### Challenges

- 1. User Involvement and Control :** Designing features to keep users engaged and coming back to the app regularly , such as daily activities , rewards , and events can be challenging to effectively.
- 2. Monetization Strategy :** Designing and implementing effective monetization

strategies such as app purchases while maintaining a positive user experience and maximizing revenue can be a challenge .

**3.Security and Privacy Protection :** Sometimes keep the security tight and under control might be too risky. Implementing robust security measures to protect user data and prevent unauthorized access or breaches can be challenging .

**4. Managing User Library :** Design an intuitive and user friendly interface that allows users to easily navigate and manage their library. Implement automatic syncing and backup features to regularly save users game progress.

**5. Simple Search and Filter Function :** Create a user friendly interface that includes search bar and filter options prominently displayed on the screen.

### **Project Goals**

Users who regularly use the app arcade can discover new apps and games.It also helps users to discover new and creative content. Developers can showcase their skills and talent to discover their apps and games to the arcade.This will encourage them to submit . User reviews and ratings can bring back users into the arcade. Also collect data about user behaviour ,app performance and optimize the app arcade.

### **Used Technologies for the project**

#### **Front End:**

HTML  
CSS  
JavaScript

#### **Back End:**

C#

#### **Database:**

SQL

#### **Platform:**

ASP.net  
Visual Studio  
MySQL

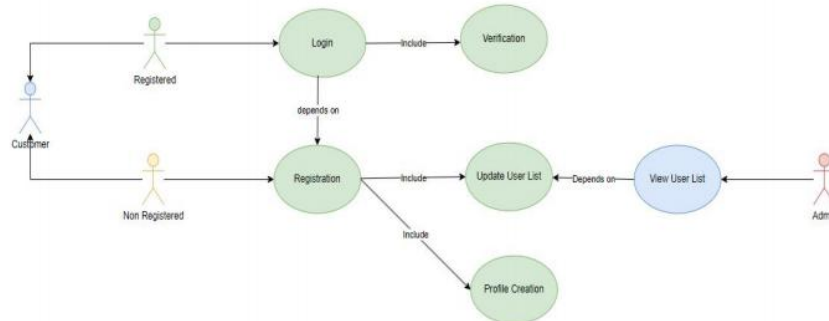
### **Use Case Diagram**

We have used some subsystems and based on the subsystem we have created some Use-Case diagram.

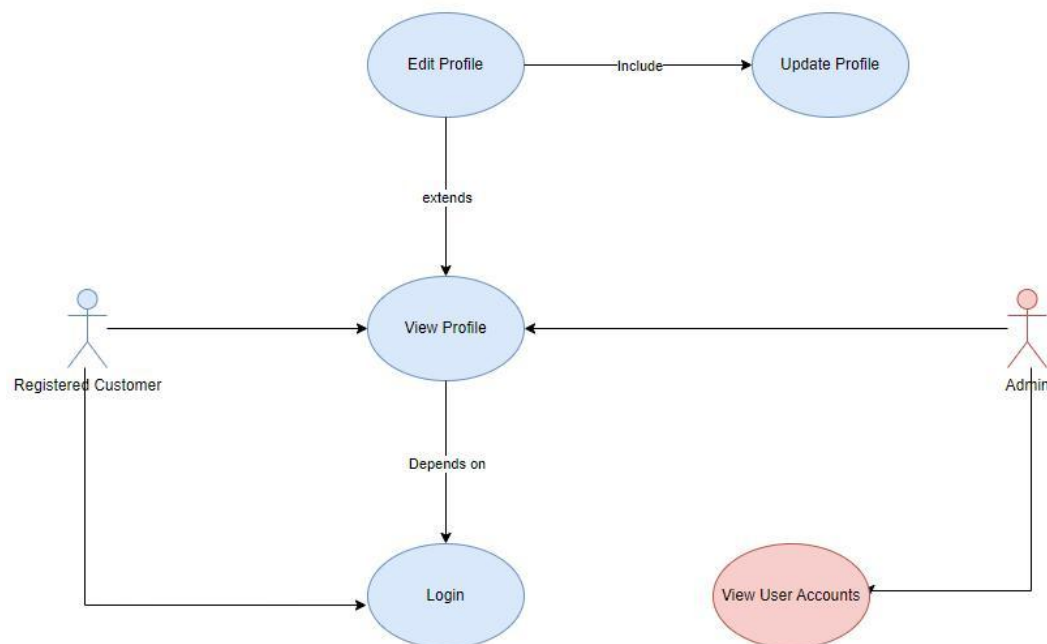
#### **Subsystems :**

**1. Login & Registration:** In login subsystem priority will be high , primary actor

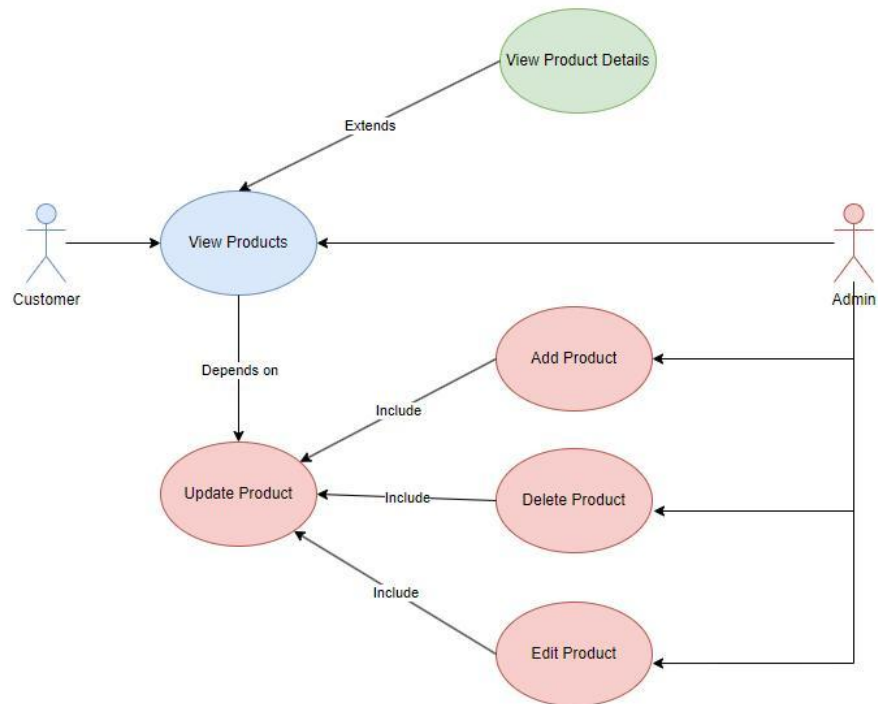
will be user and precondition is user must register first. Here is the Use-Case diagram of login subsystem-



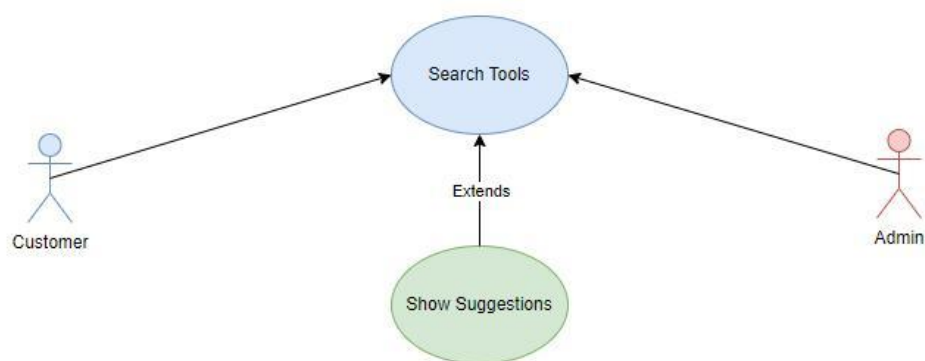
**2. Account :** In account subsystem priority will be medium , primary actor will be user , secondary actor will be admin and precondition is user must register and login. Here is the Use-Case diagram of account subsystem-



**3. Product & Search :** In search and product subsystem priority will be high , primary actor will be admin , secondary actor will be user and precondition is none. Here is the Use-Case diagram of search and product subsystem-

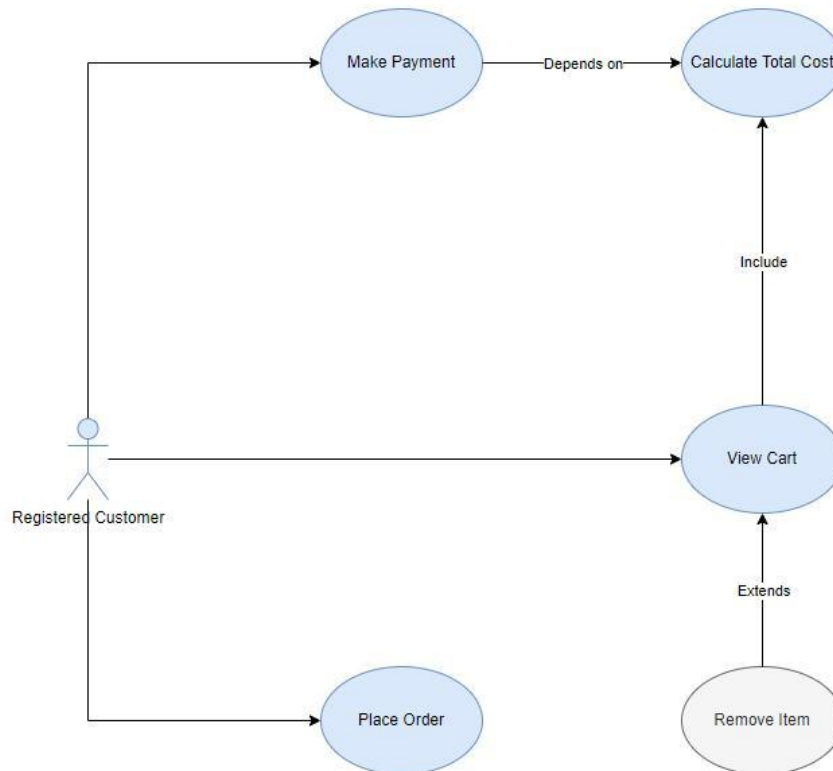


**Product Subsystem**

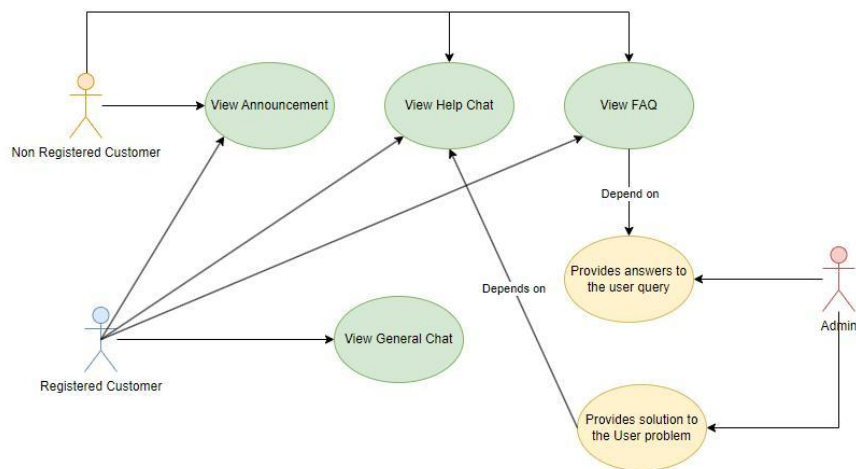


**Search Subsystem**

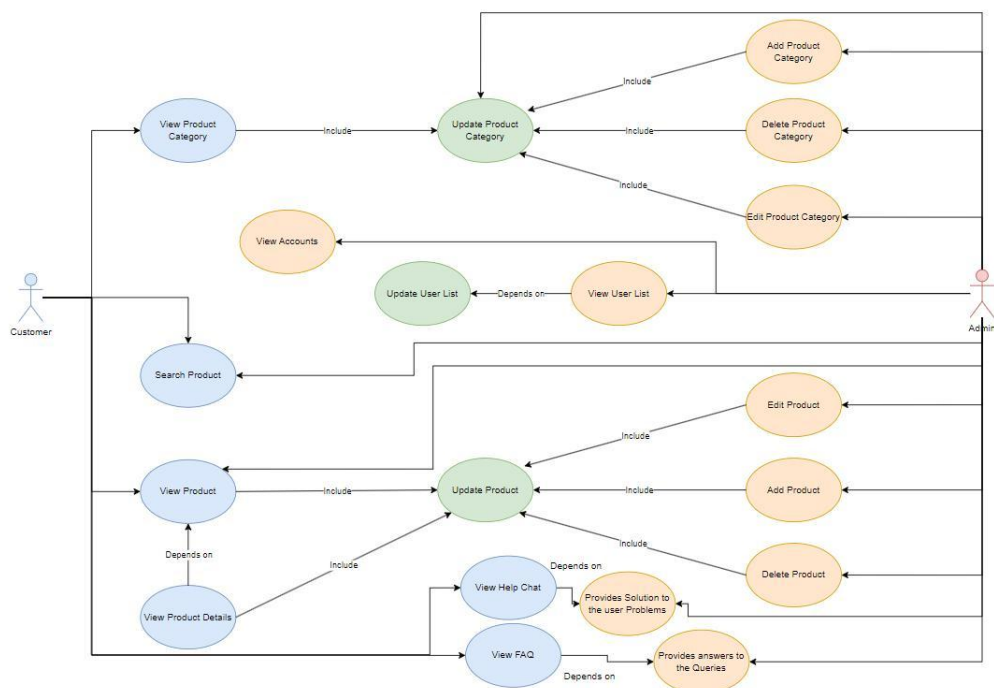
**4. Ordering :** In order subsystem priority will be high , primary actor will be admin and precondition is User must login. Here is the Use-Case order subsystem-



**5. Community Chat & Helpline :** In community chat and helpline subsystem priority will be medium , primary actor will be user , secondary actor will be admin and precondition is none. Here is the Use-Case community chat and helpline subsystem-



**6.Admin Panel :** In admin panel subsystem priority will be high , primary actor will be admin and precondition is admin must have security access. Here is the Use-Case admin panel subsystem-



## Data Flow Diagram

### **DFD ACTIVITIES OF THE PROJECT :**

#### **For Users :**

- ✓ Users can register and log in.
- ✓ Users can search apps.
- ✓ Users can purchase apps.
- ✓ Users can Chat or Communicate .

#### **For Admin :**

- ✓ Admin can maintain Database.
- ✓ Admin can manage users.
- ✓ Admin can communicate users.
- ✓ Admin can manage Products.
- ✓ Admin can update profile.

### **Main Process and Sub-process Names: :**

#### **1. Processing User Data**

##### **1.1. Login**

##### **1.2. Registration**

#### **2. Product**

##### **2.1. Add Product**

##### **2.2. Update Product**

##### **2.3. Delete Product**

#### **3. Order**

##### **3.1. Search**

##### **3.2. Payment**

##### **3.3. Select**

##### **3.4. Manage**

##### **3.5. Payment Verification**

##### **3.6. Place order**

### **Activities of Process (DFD)**

#### **Level 0**

- ✓ The Abstraction view of the whole system.

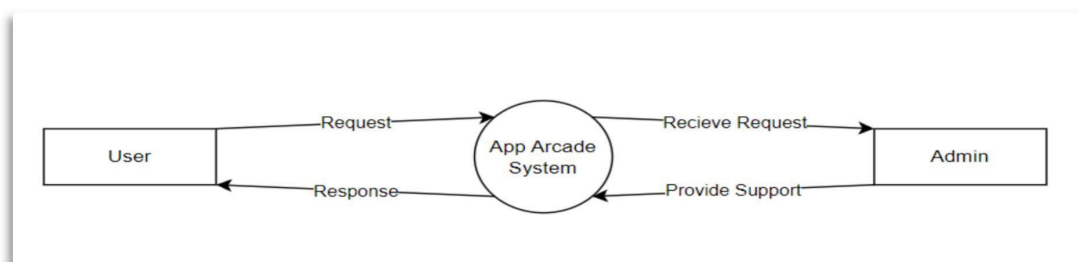
#### **Level 1**

- ✓ Highlight the main processes of the system.

#### **Level 2**

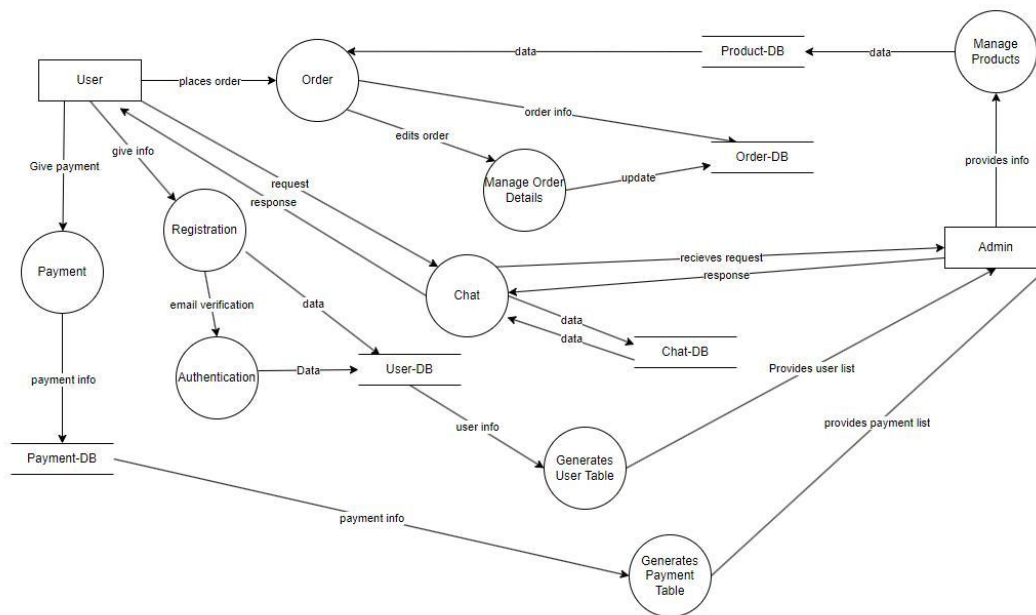
- ✓ Decompose the processes to subprocesses.

### Level 0 DFD



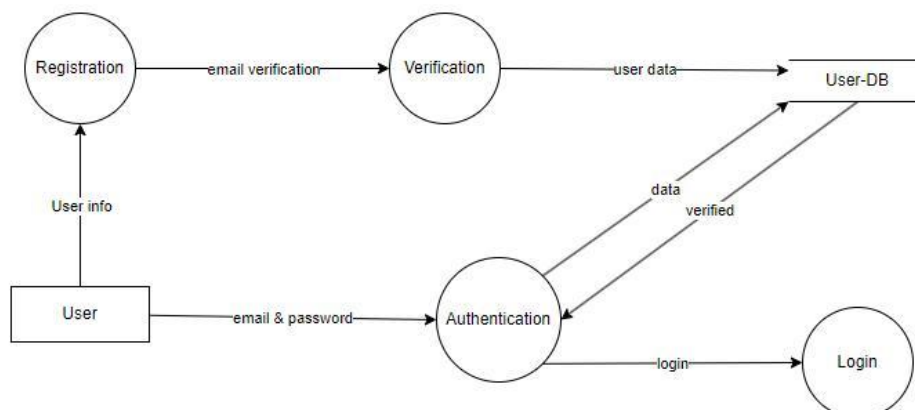
In App Arcade System User can request and Admin can receive request from users and Admin can provide support and User can response from that.

## Level 1 DFD



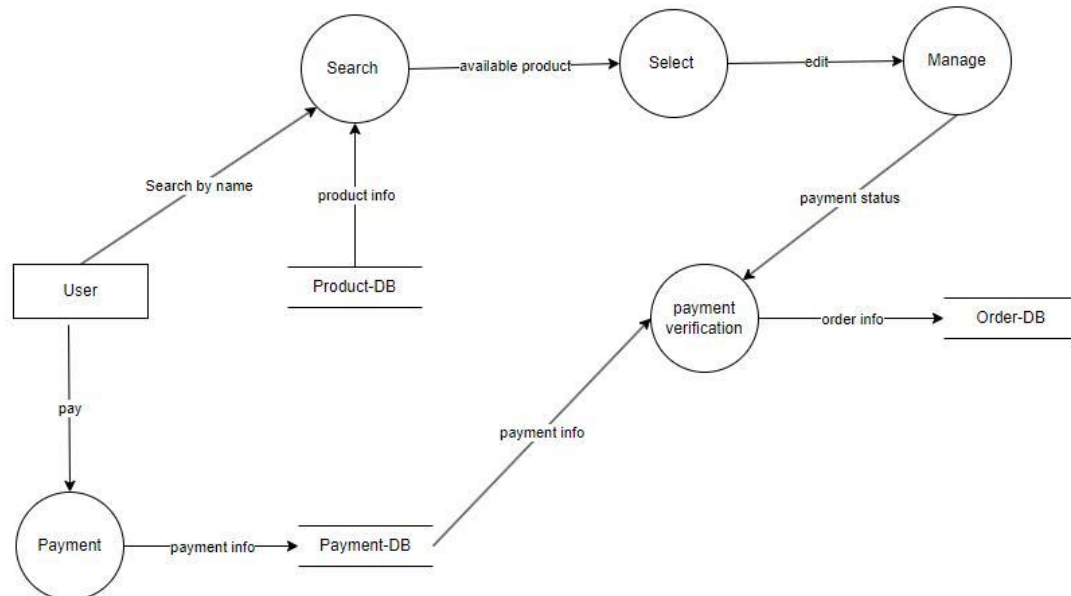
Here, User can pay through “give payment” and the “payment info” will be provided to Payment-DB. User will “give info” through Registration and after that there is an “email verification” which leads to Authentication and that “data” will be provided to User-DB. User can “request” through chat and that data will be provided to Chat-DB and from that Chat-DB will “response” through chat to Users. Users can “place order” and that “order info” will be provided to order-DB. Admin can manage products by “providing info” and that “data” will be forwarded to product-DB. Admin provides “user list” and then generates user table by following the “user info”. Admin provides “payment list” and then generates payment table by following the 3 “payment info”. Admin can receive “request” from Users and also “response” to the Users through chat. Admin can manage order details by updating order details or can edit order details.

## Level 2 DFD (Registration & Login)



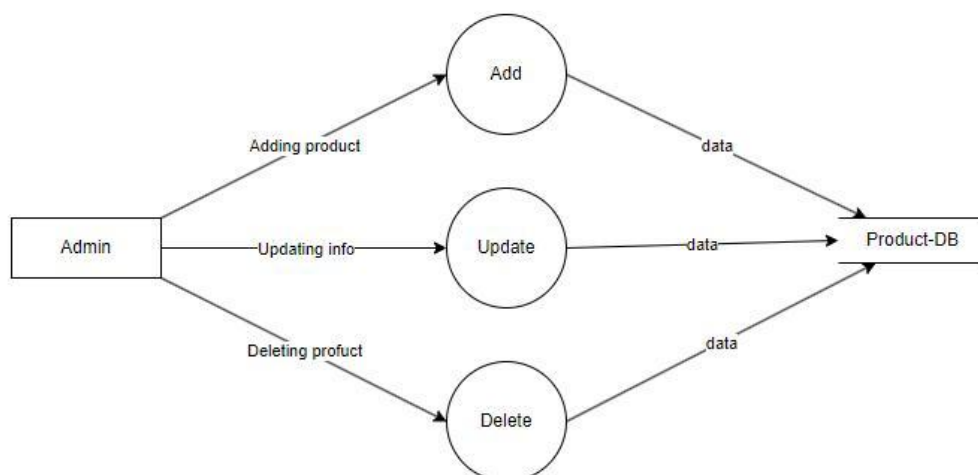
Here, User can register by giving user info and there will be email verification and that data will be provided to User-DB. Users email & password have Authentication after verifying from the User-DB and then User can login.

### **Level 2 DFD (Order)**



Here, User can pay through “give payment” and the “payment info” will be provided to Payment-DB. User can also search a product by its name and can see the product info. User can select the available product and also manage payment status according to order-DB. User will get payment verification from payment info according to payment-DB.

### **Level 2 DFD (Product)**



Admin can add product, update product, delete product and that data will be provided to product-DB.

## **Entity Relation Diagram**

### **Entity Names :**

User:

User Id (PK), Username

Email

Password

Phone number

Product:

Product\_id (PK)

Product\_name

Image

Price

Description

Category (FK)

Order:

Order\_id (PK)

UserID (FK)

Product\_ID (FK)

Payment:

Transaction\_ID (PK)

Order\_id (FK)

Category:

Category\_ID (PK)

Category\_Name

User Library:

Lib\_id(PK)

Product\_id (Fk)

Category\_id (FK)

Admin:

Admin\_id (PK)

Name

Password

Email

Chat:

Message\_ID (PK)

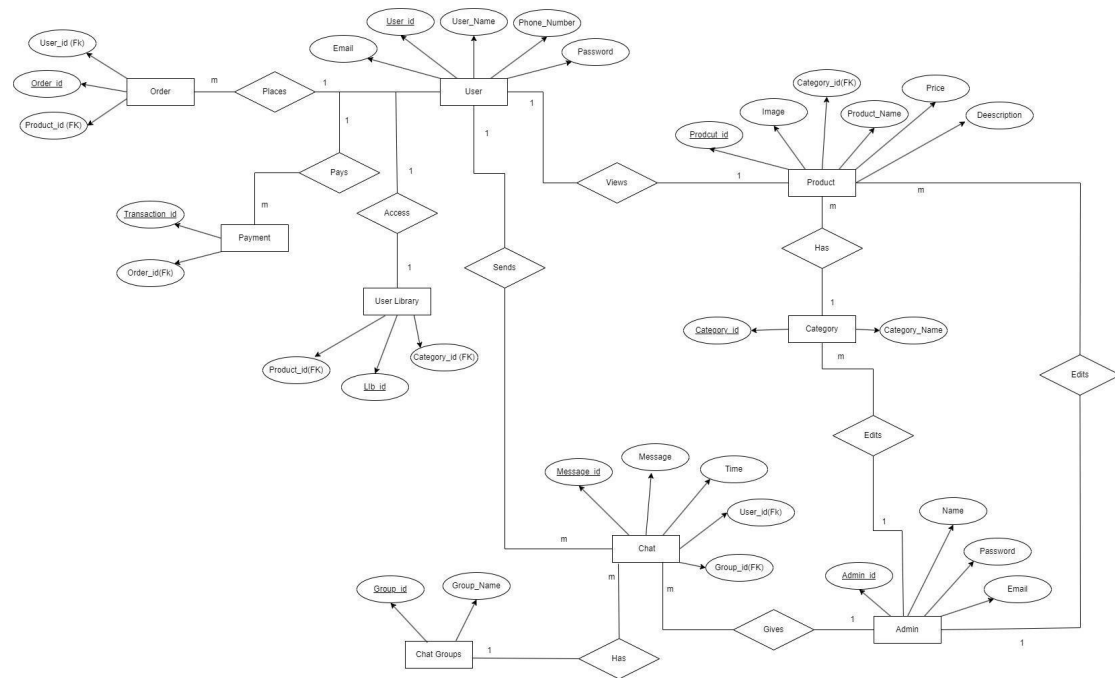
Message

Time

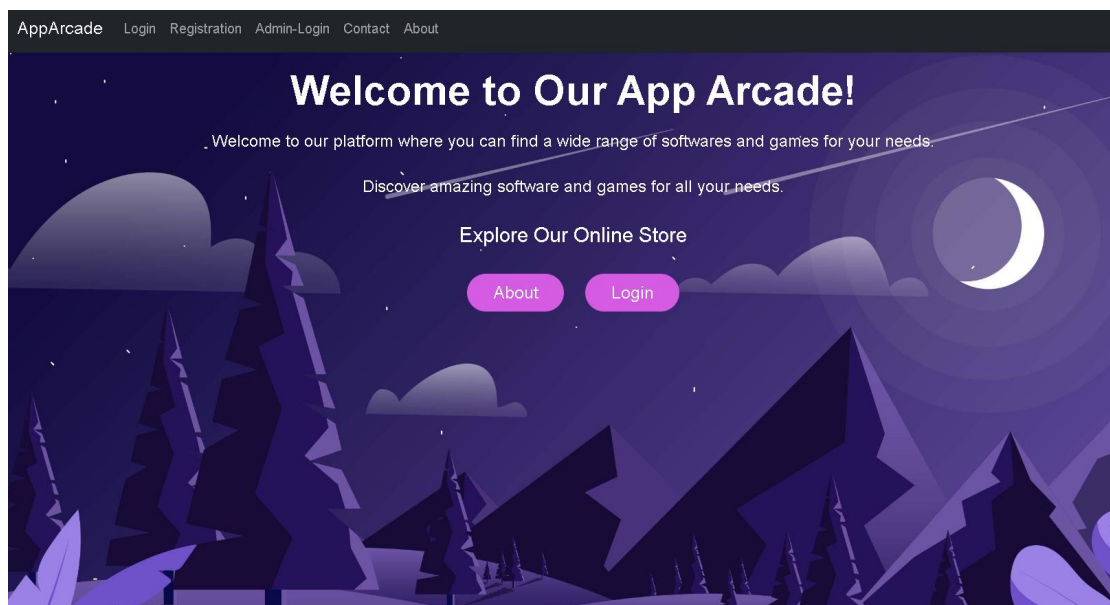
User\_id (FK)

Group\_id(FK)

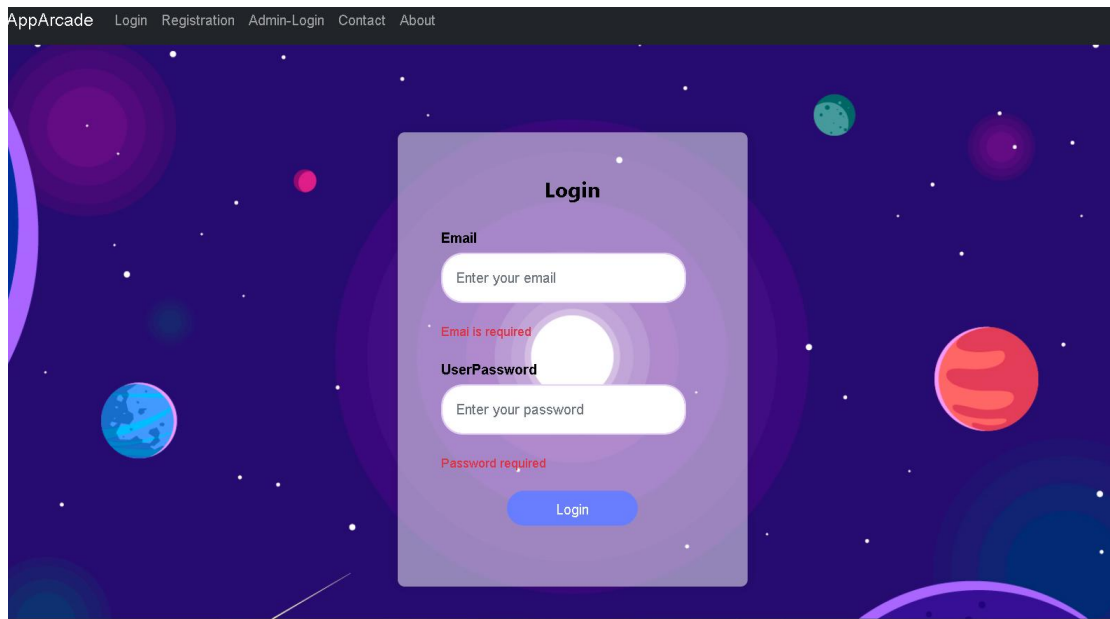
Chat Groups:  
Group\_id (PK)  
Group\_Name



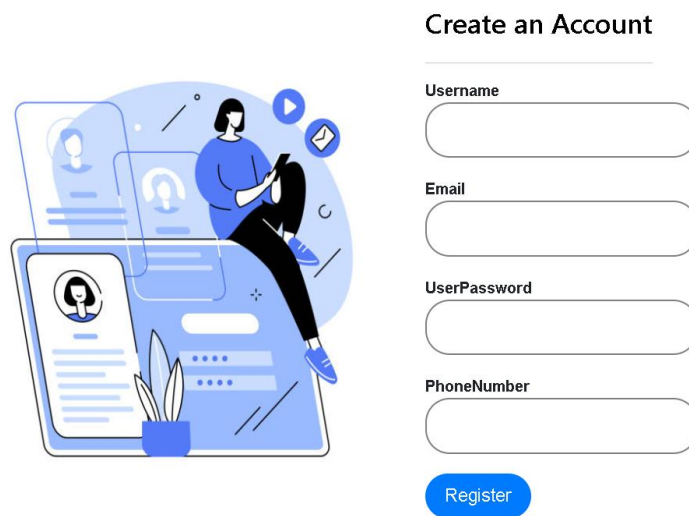
## Project Features



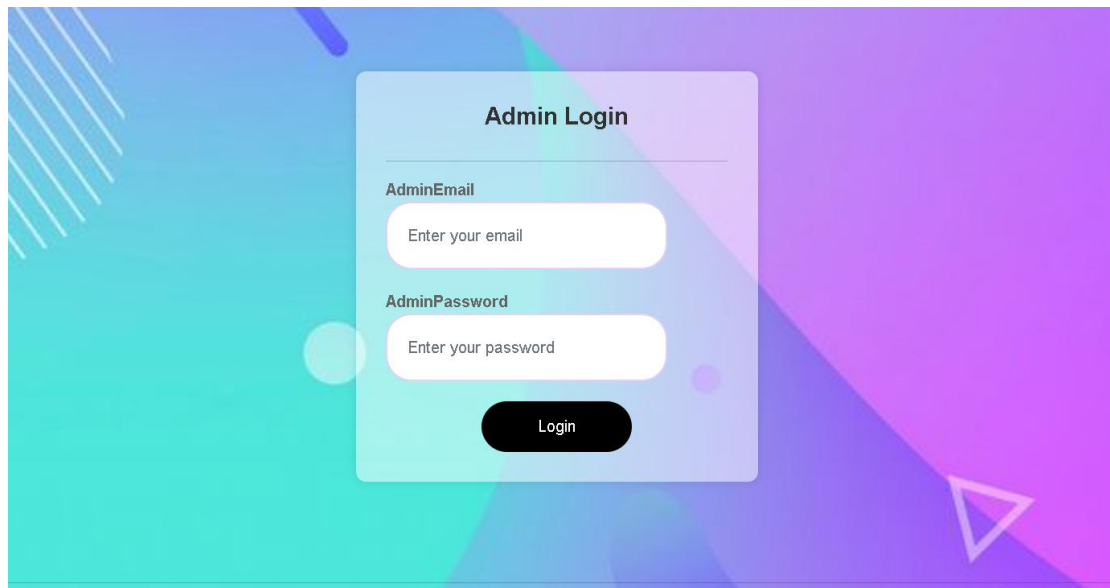
This is our main home page of our project App Arcade.



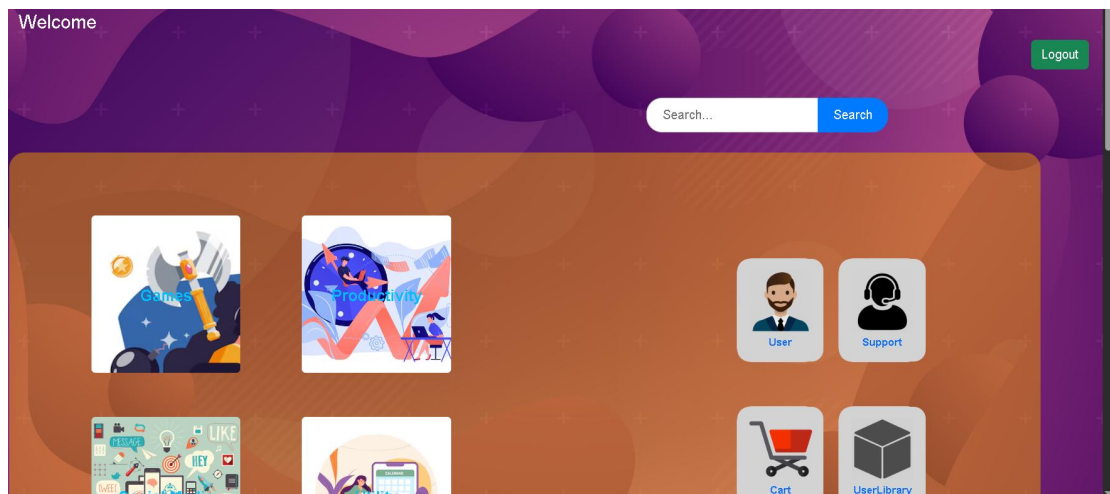
This is our Login page where registered users can access the website by providing the email and the password they registered with.



Here is the registration page where users can register and login .



Here is admin login page where admin can login and get access.



Here users can login and see this page immediately and get access to some features like cart,user library,support.

### Cart

| Item | Name | Price |
|------|------|-------|
|------|------|-------|

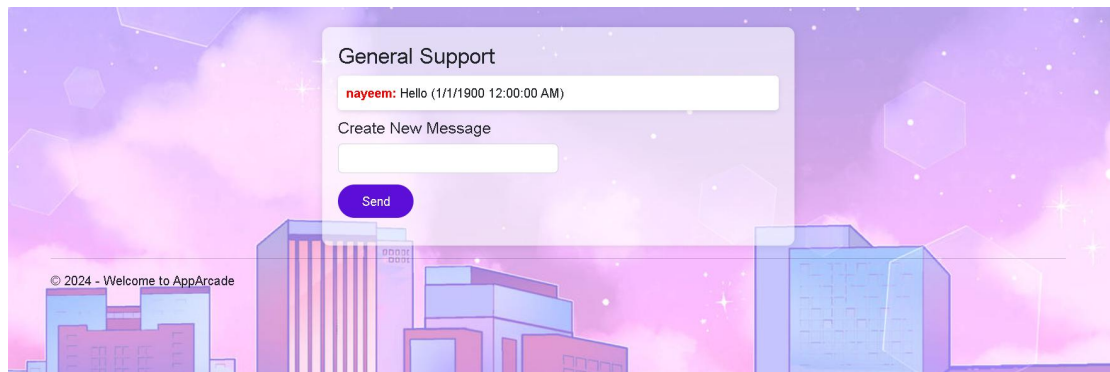
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Here the page is cart .

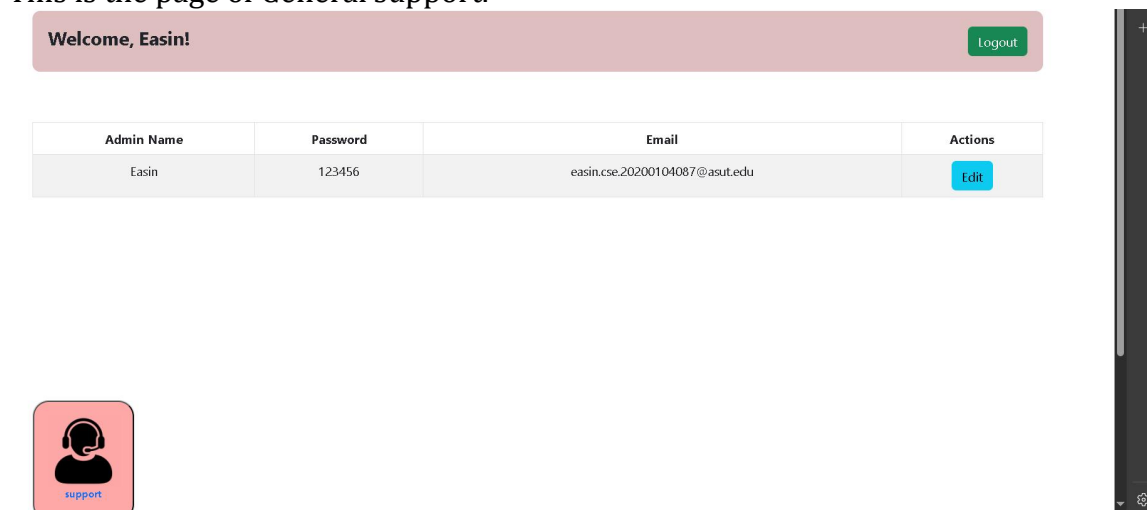
| Product_Name                                                                                 | Purchase Date                                 |
|----------------------------------------------------------------------------------------------|-----------------------------------------------|
|  Valorant | 9/22/2024 9:28:25 PM <a href="#">Download</a> |

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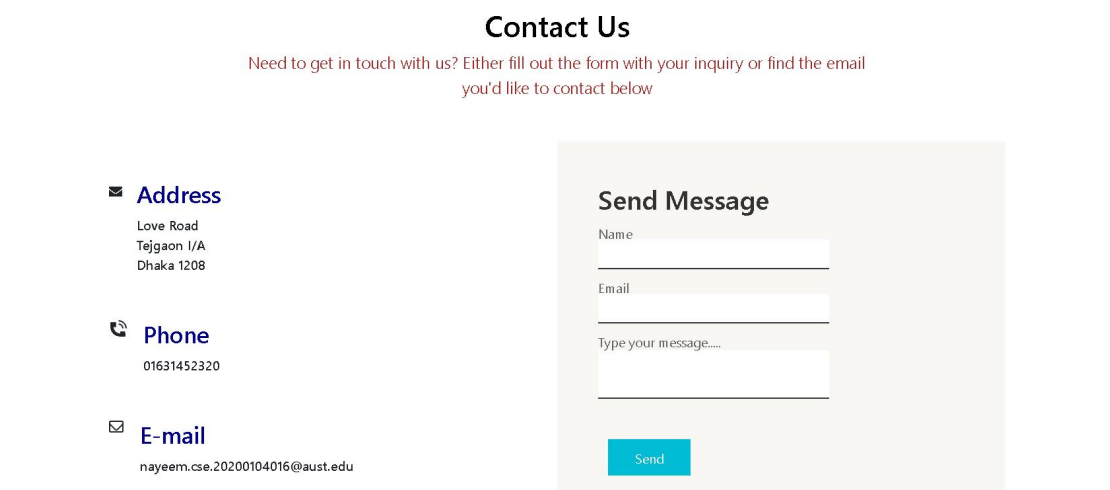
Here is the user library where users can buy product and get the details in user library.



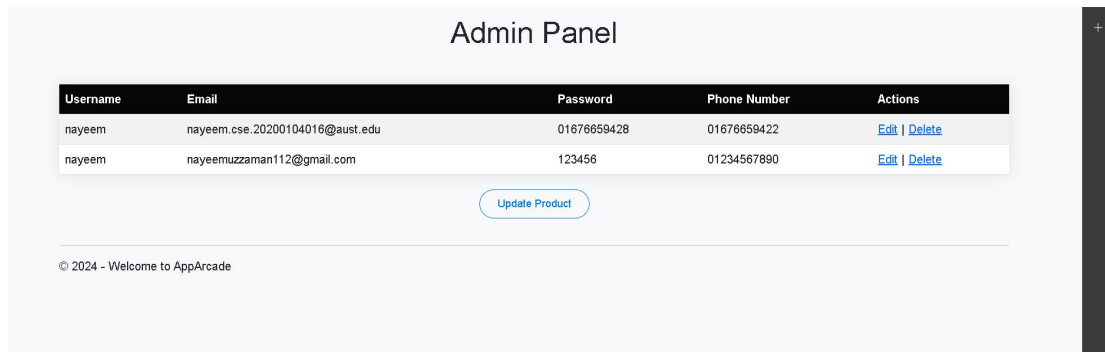
This is the page of General support.



After login admin will see this page .



Here users can contact us through email by providing their name and email address.



Here admin can update any products.

## **Opportunities of Development**

We aim to improve the project by adding further features to the website. This may include introducing an array of games and applications made for PC ,mobile devices, and gaming consoles. We also aim to introduce offering product keys or license of various software's such as Office, Matlab or Operating systems such as windows. Finally we aim to further improvise the interface and support to be more user friendly in the future.

## **Contribution**

This project was done by 2 people with = equal contributions. Research & Planning , Back End , Database was done by Easin. UI Design , Documentation , Front End was done by Nayeem . Testing & Debugging was done by both .

## **Conclusion**

App Arcade has the potential to grow into a profitable and sustainable platform if it carefully balances quality, innovation, and user experience while staying adaptable to market changes and technological advancements.

## **References**

### **1. Mobile Gaming Industry Reports**

Newzoo Global Games Market Report: Newzoo publishes annual reports on the global games market, covering industry trends, market size, and projections. It's a valuable resource for understanding the mobile gaming landscape.

### **Newzoo: Global Games Market Report**

### **2. App Arcade Competitors**

Apple Arcade: Launched by Apple in 2019, it is a subscription-based service that provides access to exclusive, ad-free games. Study Apple Arcade's business model, technical infrastructure, and game curation process.

### **Apple Arcade**

**Google Play Pass:** Google's alternative to Apple Arcade, offering a subscription model with access to premium apps and games. Its strategy can be insightful for App Arcade.

### **Google Play Pass**

**Microsoft's Xbox Game Pass:** While focused on console and PC gaming, Game Pass's subscription model and cloud gaming integration can provide insights for App Arcade's future technical and business growth.

#### **Xbox Game Pass**